

Gentle by knowing: person knowledge, not force thresholds, protects restricted joints in simulated robotic arm lifting

Mu-Hua Wang

Abstract—Many care recipients cannot move a shoulder or elbow freely, and a robot that lifts their arm can press a restricted joint into the end of its range. Robot safety standards and recent limb-moving robots rely on contact-force limits. Unfortunately, when a robot supports a multi-joint limb from outside, the loaded joint is not on the axis of its force sensor. In contrast, we propose that gentleness requires knowing the person before contact and being willing to decline, and vary that knowledge. In a musculoskeletal simulation of a robot lifting the forearm from below, controllers know nothing (a fixed rule), a categorical flag, recorded joint ranges with 20° error (used by a population body model or by a policy learned offline in simulation), or everything (an oracle). A *harmful outcome* is a robot-caused end-range joint load of at least 0.5 N·m, a mechanical-exposure proxy; a *safe completion* is a completed lift below it. Hypotheses were written in a dated protocol before each round and tested on new simulated persons. No force-magnitude guard reduces harmful outcomes by significantly more than 5 percentage points: the guards prevent at most 13% of the fixed rule’s harmful outcomes, often do not trip before the joint is loaded, and the fixed rule’s harmful lifts stay within the ISO/TS 15066 contact-force limit for the lower arm. With a complete record, the learned policy reduces harmful outcomes from 110 to 25 of 300 persons by declining 91 of them (safe completions 174 vs. 177; non-inferiority not established); at a 2.5 N·m harm threshold it completes 6–8 percentage points fewer lifts than the fixed rule. The reduction replicates and survives posture noise, over-estimated records, a shifted population with more and tighter restrictions, graded end-range resistance and two physics perturbations, but shrinks by almost half when half of the restrictions are unrecorded. Post hoc, knowing which joint is restricted, plus the settled posture, suffices. Code, protocol and per-episode data are public.

Index Terms—Physical human–robot interaction, caregiving robots, gentle handling, safety, musculoskeletal simulation, joint contracture, range of motion, abstinence

I. INTRODUCTION

Lifting a person’s arm—to wash the axilla, pass a sleeve or reposition the person—is among the most frequent actions in care and a target for assistive robots [1]–[4]. Doing it gently matters because many care recipients cannot move freely. About half of adults admitted with stroke develop at least one contracture within six months [5], and contractures are common in long-term care [6]. Severe spastic contractures independently predict minimal-trauma fractures in long-term care residents [7]; in people with severe motor and intellectual disabilities, some fractures occur during diaper changes,

repositioning or transfers [8]; and in hemiplegia, the way the arm is moved is associated with shoulder pain [9], [10].

Collaborative-robot safety bounds contact force and pressure with limits derived from pain-onset thresholds [11], and injury knowledge enters robot control for impacts [12]. Adjacent standards go further: wearable assistive robots apply assist force only within the user’s passive range of motion [13], rehabilitation robots limit movement to pre-set, patient-specific limits [14], [15], and personal-care robots seek external assistance or stop when decision uncertainty remains unacceptable [16]. None of these states what a robot that moves a limb by end-effector contact must know about the person, or whether sensing contact force is enough.

Our key insight is that, when a robot supports a multi-joint limb from outside, the joint it loads is not on the axis of its force sensor. A joint load that matters is only a few newtons at the contact point, hidden under the limb’s own weight, so a force guard trips only after the joint is loaded (Figure 1C). A robot therefore cannot feel its way to gentleness; it must know the person before contact and be willing not to lift. Human carers work this way: they read in the chart that “the left shoulder is stiff” and handle the arm differently or ask for help.

We test two hypotheses with a simulated robot that lifts the forearm of persons with restricted shoulders and elbows (Figure 1): force-magnitude guards do not reduce harmful outcomes by significantly more than 5 percentage points, and knowledge of the person, with the option to decline, reduces them without losing safe completions.

Related work

Robots that move limbs. Assistive systems already respect force or joint limits given in advance. Simulation frameworks model care recipients with physical limitations [17], [18] and train assistive policies for simulated impairments [19]. Physical systems lift a bed-bound person’s limb within contact-force bounds [2], reposition limbs within age-group joint limits [3], plan dressing within each participant’s measured range [20] and predict functional range from clinical scores [21]. None measures harm across levels of information about the person or compares that information with force sensing.

Force limits and joints. The same external force can correspond to very different joint loads [22]; a robot’s wrist force sensor feels the rendered end-range resistance of a haptic shoulder device [23]; and industrial force limits do not transfer

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M.-H. Wang is with the Department of Biomedical Engineering, National Yang Ming Chiao Tung University, Taipei, Taiwan (e-mail: may.be13@nycu.edu.tw).

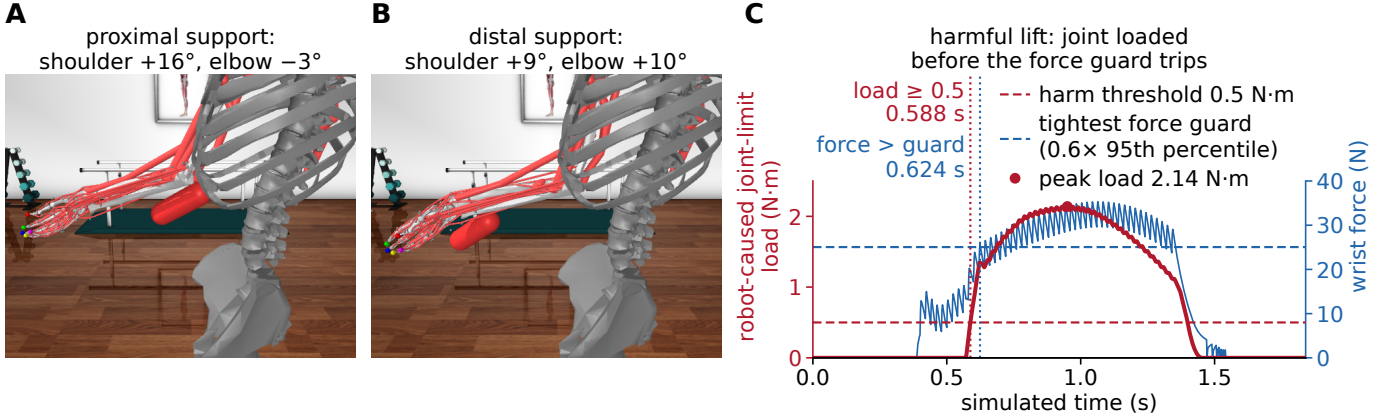


Fig. 1. The robot loads the joint before its force sensor notices. (A, B) A capsule supports the forearm from below and raises it 5 cm; proximal support elevates mainly the shoulder, distal support also flexes the elbow (joint excursions for one unrestricted simulated person, seed 104), so for a person with a restricted shoulder or elbow the wrong site presses that joint into the end of its range. (C) One harmful lift of the fixed rule (seed 2222, restricted elbow): the robot-caused joint-limit load passes 0.5 N·m at 0.588 s, before the wrist force exceeds the tightest force-guard threshold (25.05 N, 0.6 times the 95th percentile of harmless force) at 0.624 s.

directly to patients, although force control helps keep joints in range [24]. In joint-aligned single-degree-of-freedom devices, torque thresholds find the end of range without reported safety problems [25], [26]. Force sensing thus works when the sensor sits on the joint's axis; in end-effector support of a multi-joint limb it does not.

Value of information. Declining is the reject option of classification [27], and we measure the value of information [28] empirically. Comparing a baseline, an inference model and an omniscient planner has precedent in autonomous driving [29], and assistive robotics compares generic and personalised human models [30]–[32]. To our knowledge, no work varies the person-specific body information available to a physically assistive robot and reports harm and safe completion at each level, with declining as an option.

Contributions

Our contributions are:

- 1) Evidence that force-magnitude guards do not substitute for knowledge of the person (Figure 2).
- 2) An *information ladder* over six levels of knowledge (Figure 3), in which a policy learned offline in simulation cuts harmful outcomes to about a quarter of a fixed rule's when the record is complete, survives several degradations, and loses much of its benefit when the record misses restrictions (Figure 4).
- 3) A post hoc analysis showing that the robot needs to know which joint is restricted, not by how many degrees, and an account of what does not help, with every hypothesis reported (Supplementary Table S1).

II. MATERIALS AND METHODS

A. Simulated body and population

The simulation uses the MyoSuite 2.12.2 myoArm model [33] in MuJoCo 3.6 [34], trunk upright, with the humerus free of the thorax so that the arm hangs from the shoulder complex. A stretch reflex with muscle velocity and length

feedback drives the muscles, following Yu et al. [35], with a 25-ms afferent delay (our choice) and a tonic baseline activation of 0.01; the simulated person neither assists nor resists voluntarily. Muscle strength, passive muscle force and joint damping vary log-normally across persons ($\sigma = 0.25$; segment masses and inertias $\sigma = 0.175$; Supplementary Table S3).

Each person independently has a restricted shoulder elevation with probability 0.4 and a restricted elbow flexion with probability 0.3, so 58% have at least one restricted joint, comparable to 52% after stroke [5] and almost two thirds in long-term care [6]; the per-joint rates and small free ranges represent a severely affected group. A restricted joint moves a free range drawn from $U(4^\circ, 16^\circ)$ (shoulder) or $U(0^\circ, 10^\circ)$ (elbow) above the person's settled posture before its limit engages. As stated in the protocol, the free ranges overlap the task's joint excursions (roughly 7° – 16° at the shoulder and up to 10° at the elbow; Figure 1), so absolute harm rates reflect the population design and only paired differences between controllers are interpreted. A *shifted* population has restriction probabilities 0.6 (shoulder, $U(2^\circ, 12^\circ)$) and 0.5 (elbow, $U(0^\circ, 8^\circ)$). The joint limit is hard by default; under *graded end-range resistance*, resistance ramps up over a window of $w \in \{6^\circ, 12^\circ, 20^\circ, 25^\circ\}$ centred on the limit (our derivation from [36]–[38]).

B. Robot and task

The end effector is a position-servoed capsule with three translational and three rotational joints (effectively five degrees of freedom; 6000 N/m, 20 N·m/rad, 50 Hz). It approaches the forearm from below, raises the forearm centre by 5 cm in closed loop on the observed forearm height, and holds for 0.5 s. A *strategy* is a support site (8 sites, 3 cm apart), an approach angle (-40° , -20° , 0°) and a speed (0.10 or 0.25 m/s); 48 strategies. No single strategy is safe for everyone (Figure 1).

C. Gentleness: harmful outcomes and safe completions

Harm is the load the robot presses into the shoulder-elevation and elbow-flexion limits: the peak 50-ms moving average of the joint-limit constraint torque (excluding the first 100 ms), summed over the two joints; from round 23 onward, limit torque present before contact is subtracted. An outcome is *harmful* if $\text{harm} \geq 0.5 \text{ N}\cdot\text{m}$ (primary threshold). We also report 2.5 N·m, near the mean torque ($2.7 \pm 1.1 \text{ N}\cdot\text{m}$) that therapists deliberately apply to reach passive elbow end range in children after fracture [39], a threshold sweep and a threshold-free mean load. Such loads are routine in supervised assessment; the thresholds therefore mark unintended, unmonitored end-range loading by a robot—a mechanical-exposure proxy, not injury. A *safe completion* is a completed lift with harm below the threshold; a declined lift counts as neither. No controller except the oracle sees harm.

D. What the robot may know

The *care record* gives each joint’s free range with Gaussian error of $\text{SD } \varepsilon \in \{4^\circ, 10^\circ, 20^\circ\}$; inter-rater standard errors of $14\text{--}25^\circ$ for symptomatic shoulders [40] and $6\text{--}10^\circ$ within a session after stroke [41] make 20° the realistic default. Unrestricted joints are recorded as unrestricted, so a *complete* record always tells which joints are restricted—an idealisation. Real records are often categorical, such as the MDS 3.0 item that codes shoulder, elbow, wrist and hand together [42], and incomplete: in one nursing-home study, reduced range was found in 69% of residents but documented for 4% [43]. We also use a categorical flag (shoulder and elbow merged) and a *half-documented* record that omits each restricted person with probability 0.5. The robot observes the settled shoulder and elbow angles before contact and its wrist force during the lift.

E. Controllers

Controllers differ only in what they know and whether they may decline.

- **F (fixed rule).** Among strategies completing $\geq 90\%$ of 60 training persons, the one with the lowest median harm.
- **R (force-magnitude guard).** Runs F; if the wrist-force magnitude exceeds m times the 95th percentile of harmless training lifts ($m \in \{0.6, 0.8, 1.0, 1.2\}$), it stops, retracts and tries the best strategy supporting at least 9 cm away, up to three attempts.
- **CAT, CAT_{half}.** Decline if the complete or half-documented flag marks a restriction; otherwise F.
- **S (record threshold).** Decline if a recorded free range is below 3.5° (locked on 280 earlier persons); otherwise F.
- **M (population body model).** A library of 199 simulated unrestricted bodies predicts each joint’s excursion and each strategy’s completion. M keeps F if F is predicted to stay 3° inside both recorded ranges; otherwise it uses the margin-keeping strategy with the highest predicted completion (≥ 0.8); otherwise it executes the least-overshooting strategy if the overshoot is at most 3° , and otherwise declines.
- **L (policy learned offline in simulation).** From 400 simulated training persons with outcomes for all 48 strategies,

a k -nearest-neighbour model ($K = 25$) over six standardised features—two recorded free ranges (clipped at 30°), two unrestricted flags and two settled joint angles—estimates each strategy’s probability of a safe completion. L picks the best strategy and declines if its estimate is below $\tau = 0.65$; the training-set value with the fewest harmful outcomes among those keeping safe completions within 2.5 percentage points of F, selected with each training person among its own neighbours (Section III-B reports a leave-one-out re-selection). Persons recorded as unrestricted receive F. A four-feature variant without posture ($\tau = 0.60$) was specified in advance for round 25.

- **O (oracle).** Knows every outcome; picks a safe completion if one exists, otherwise declines.

F. Protocol and statistics

The protocol document began as the plan of an earlier phase (rounds 1–16, other tasks; main hypotheses not supported); rounds 17–26 (Supplementary Table S6; 18R is the frozen replication of round 18) were added as dated amendments, each written before that round’s data were generated. It was kept locally, not externally registered. Every confirmation set consists of simulated persons never used before; earlier confirmation persons were later reused for development and training.

All comparisons are paired. Differences in harmful-outcome and safe-completion rates carry 95% confidence intervals (paired t ; the analysis code uses $t = 2.1$ for $n > 20$, slightly conservative). A harm hypothesis is supported if the upper bound is below zero (for force guards, below -5 percentage points); non-inferiority uses margins fixed in advance ($2\text{--}2.5$ points, or a stated count), equivalence ± 0.05 . Differences are proportions ($-0.19 = 19$ percentage points fewer). The original plan prescribed Holm–Bonferroni correction for the earlier phase; rounds 17–26 are not corrected for multiplicity. Post hoc analyses tune baselines on the 400 training persons only and evaluate them on the 600 persons of rounds 24–25 with complete outcome grids. Supplementary Table S2 lists the deviations: a round-17 bug found after data generation, a harm definition corrected after round 22, the change of primary outcome after round 18R, testing L’s safe completions only against M after its non-inferiority to F failed in round 24, and three audit checks written after rounds 24–25 had been evaluated.

III. RESULTS

A. Force-magnitude guards do not substitute for knowledge

The fixed rule F harms 34–40% of persons in every standard-population hard-limit set (e.g. 229 of 600 in round 22). No force-magnitude guard reduces the harmful-outcome rate by significantly more than the prespecified 5 percentage points, at any of four thresholds, under a hard limit or graded resistance of 6, 12, 20 or 25° (Figure 2 shows the hard limit and $12\text{--}25^\circ$). This is a failure to show a large reduction, not proof of none: the tightest guard reduces harm slightly but significantly at 6° (-4 points), 20° (-4) and 25° (-7 [-11 ,

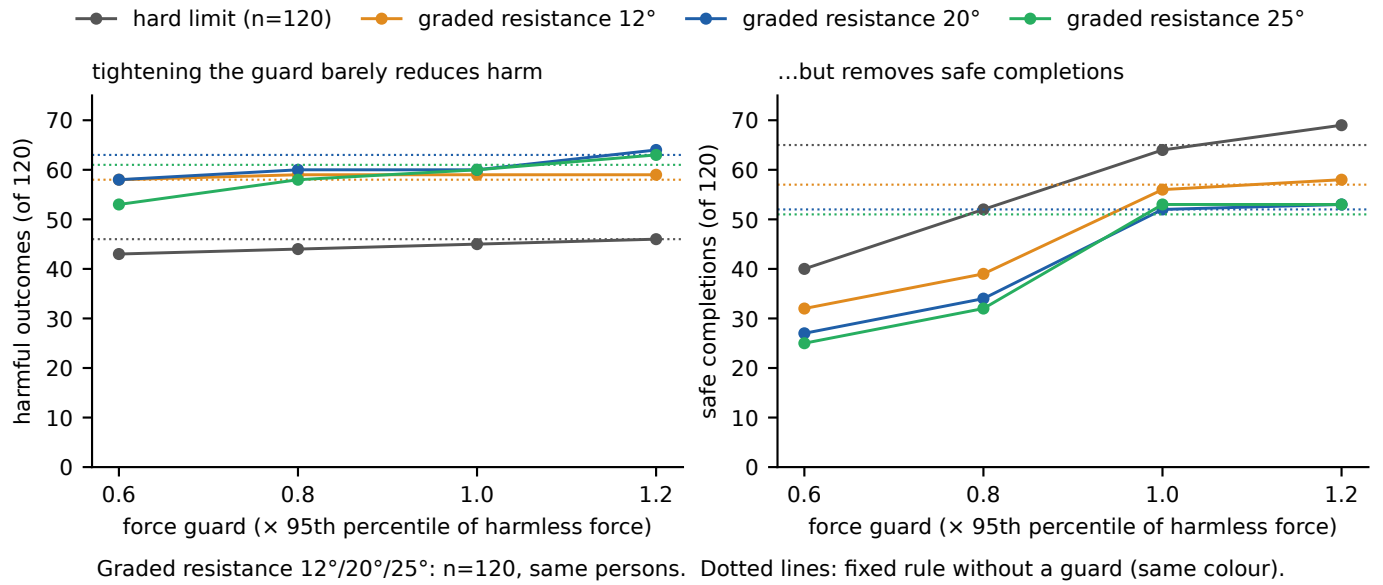


Fig. 2. Tightening a force-magnitude guard removes safe completions but barely changes harmful outcomes. Four guard thresholds under a hard limit (round 21, 120 new persons) and graded end-range resistance of 12, 20 and 25° (round 23, the same 120 new persons under each width); left, harmful outcomes; right, safe completions. Dotted: fixed rule without a guard.

–2]), not with a hard limit or at 12°, each time at the cost of 40–51% of safe completions.

Between 95% and 100% of harmful outcomes occur during the first attempt. Post hoc, each guard prevents only 0–13% of F’s harmful outcomes (0–10% with a hard limit), mostly by aborting the lift; in the remaining harmful first attempts the guard never trips in 33–79% (hard limit) and 42–95% (graded resistance) of cases, and when it trips, the joint load has already reached 0.5 N·m. Wrist force tracks joint load closely within a person only with a 12° window (median Spearman 0.75; 0.10–0.35 with a hard limit or 6°). Among the 600 persons of rounds 24–25, every harmful lift of F stays below the ISO/TS 15066 quasi-static force limit for the lower arm (160 N; simulated peak contact force median 34 N, maximum 154 N; post hoc).

B. The information ladder

Knowledge helps mainly by telling the robot whom not to lift. The body-model rule M reduces the harmful-outcome rate by 14–26 percentage points in every standard-population condition with a complete record (rounds 21–26; e.g. –0.19 [–0.23, –0.16], 113 vs. 229 of 600, safe completions 351 vs. 344), by 9 points with half of the restrictions unrecorded and by 37 in the shifted population; without its decline option it is not shown to reduce harm (Section III-E).

Figure 3 shows 300 new persons at six levels of knowledge, each paired with a decision rule (counts in Supplementary Table S4). Declining every flagged person removes all harm but 37% of safe completions (112 safe, 0 harmful); with half the flags missing it leaves 60 harmful outcomes. Ranges with 20° error plus the record threshold S or the body model M reduce harm by 39–54%. The learned policy L reduces harmful outcomes to 25, fewer than M (–0.14 [–0.19, –0.09]) with more safe completions than M (+0.04 [0.00, +0.08]); against

F it completes 174 vs. 177 lifts safely, and prespecified non-inferiority is not established (–0.01 [–0.05, +0.03]). Compared with declining every flagged person, L completes 62 more lifts safely at the cost of 25 harmful outcomes. Record accuracy matters little once L is used (4°: 184 safe, 20 harmful), and on the same persons a policy learned from the complete flag and posture alone reaches 183 safe and 25 harmful (post hoc; hollow marker): the flag rung is limited by its decline-everyone rule, not by the knowledge.

Of the 63 safe completions L misses relative to the oracle (237 safe, 0 harmful), 47 are unnecessary refusals; 18 of L’s 25 harmful outcomes are persons with no safe strategy, whom L should decline. Mean robot-caused end-range load is 6.73 N·m under F, 3.19 under M and 0.79 under L (L – F: –5.94 [–7.78, –4.10]). The advantage is not only lifting fewer people than F: post hoc, on the 423 of 600 persons L lifts, L harms 53 and F would harm 95 (–0.10 [–0.13, –0.07]). Re-selecting τ with leave-one-out (post hoc) gives $\tau = 0.60$, a different point on the same trade-off, at which L has 378 safe and 73 harmful outcomes on these 600 persons (prespecified τ : 341 and 53; F: 348 and 218).

C. Robustness of the learned policy

On further new persons (Figure 4; Supplementary Table S5), L replicates its reduction relative to F (–0.27 [–0.32, –0.21]) and to M (–0.08 [–0.12, –0.03]), but its safe-completion non-inferiority to M does not replicate (–0.01 [–0.05, +0.03]). Relative to M, the reduction holds with Gaussian posture noise (SD 5°) or without posture input; relative to F and M, it holds in the shifted population (170 vs. 38 harmful of 250, without retraining) and under 20° graded resistance. L’s safe-completion rate relative to F is –0.01 [–0.06, +0.04] in the replication, +0.02 [–0.03, +0.07] in the shifted population and +0.06 [0.00, +0.12] under 20° graded resistance.

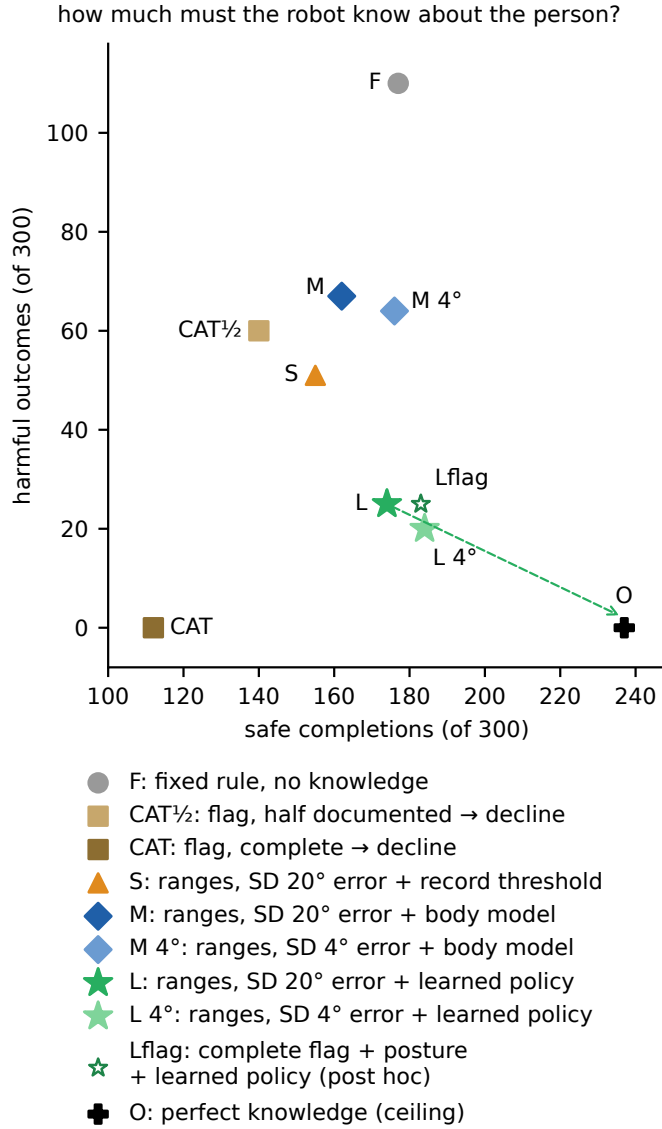


Fig. 3. Knowing the person reduces harm, mostly by deciding whom not to lift. Harmful outcomes and safe completions of 300 new persons (round 24, harmful ≥ 0.5 N·m; lower right is better) for controllers at six levels of knowledge: none (fixed rule F), half-documented flag, complete flag, ranges with 20° error (S, M, L), ranges with 4° error (M, L) and perfect knowledge (oracle O). The hollow marker (post hoc) is a policy learned from the complete flag and the settled posture only. The dashed arrow is the remaining gap from the learned policy to perfect knowledge. Counts, including declined persons, are in Supplementary Table S4.

Record completeness matters more than record accuracy. With half of the restrictions unrecorded, L's reduction relative to F shrinks from -0.27 to -0.15 (64 vs. 108 harmful), and CAT_{half} harms fewer persons than L (50 vs. 64; $+0.05$ [$+0.02$, $+0.07$]) at 29 fewer safe completions. Unrecorded persons are harmed at exactly the fixed-rule rate (50 vs. 50 of 85). With documentation as sparse as reported [43], little benefit would remain without a screen before robot use.

With three-fold stretch-reflex gains or individual variation widened from 0.25 to $\sigma = 0.40$, L's reduction relative to F and M holds, with safe completions 3.3 and 0.3 points below F's. These perturbations keep the joint-limit geometry, and F's

harm rate barely moves (36% and 40%), so they test muscle dynamics and body variation, not body structure.

Three audit checks on the 600 persons support the result. The best of all 48 fixed strategies in hindsight harms 203 persons (with only 135 safe completions), the prespecified F 218 and L 53; on persons L lifts, glenohumeral shear does not increase (paired mean -3.6 N); and when records over-estimate restricted free ranges by 5° or 10°, L's harmful outcomes rise from 53 to 57 and 68, M's from 118 to 160 and 210. In a post hoc threshold sweep (0.25–5 N·m; Supplementary Figure S1), L has the fewest harmful outcomes among F, S, M and L at every threshold, but F completes more lifts safely than L at almost every threshold; at the prespecified secondary 2.5 N·m, L completes 4–8 points fewer lifts than F (6–8 with a complete record).

D. What the robot needs to know (post hoc)

With 20° error, recorded degrees say little about a free range of a few degrees. On the 600 persons, a nearest-neighbour policy given only the two restriction flags and the settled posture ($\tau = 0.65$, also under leave-one-out) performs like L: 353 safe and 46 harmful vs. 341 and 53 (L – flags: harmful $+0.01$ [-0.01 , $+0.03$], safe -0.02 [-0.04 , 0.00]). A lookup table of the best training strategy per restriction type has only four operating points, too few to match L's trade-off; M re-tuned for 20° records harms 113 persons at 335 safe completions (L – M: -0.10 [-0.13 , -0.07]), and S cannot meet the completion criterion. Across the trade-off, the learned policies lie below the record-threshold and body-model curves (Supplementary Figure S2).

E. What does not help

Choosing the support site without declining is not shown to reduce harm in round 21 (-0.03 [-0.09 , $+0.02$]; prespecified null), although post hoc, on the 478 persons M lifts, its site choice reduces harmful outcomes modestly (118 vs. 140). Identifying the body by two gentle probe lifts is supported in round 18 ($n = 40$) but not in its replication (120 persons); in round 19 the probes load a restricted joint beyond 0.5 N·m in 18 of 71 persons, and probing harms more persons than the same rule without it (29 vs. 21 of 120). In round 19, the body model with declining harms fewer persons than the force guard (21 vs. 42 of 120) but completes fewer lifts safely (66 vs. 76).

IV. DISCUSSION

Why knowledge, and not force

In end-effector support of a multi-joint limb, the loaded joint is not the one the force sensor sees. In the simulated geometry, 0.5 N·m at the shoulder or elbow corresponds to only about 1–3 N at the contact point, small against a forearm-and-hand weight of about 17 N and guard thresholds of 25–50 N (post hoc, one person). A force-magnitude guard therefore trips late, if at all. Knowledge acts before contact, where the decision that matters most is whether to lift at all. Force limits remain necessary for contact safety but are not sufficient for joints. Declining echoes robots that ask for help under ambiguity [44], here triggered by predicted harm to the body.

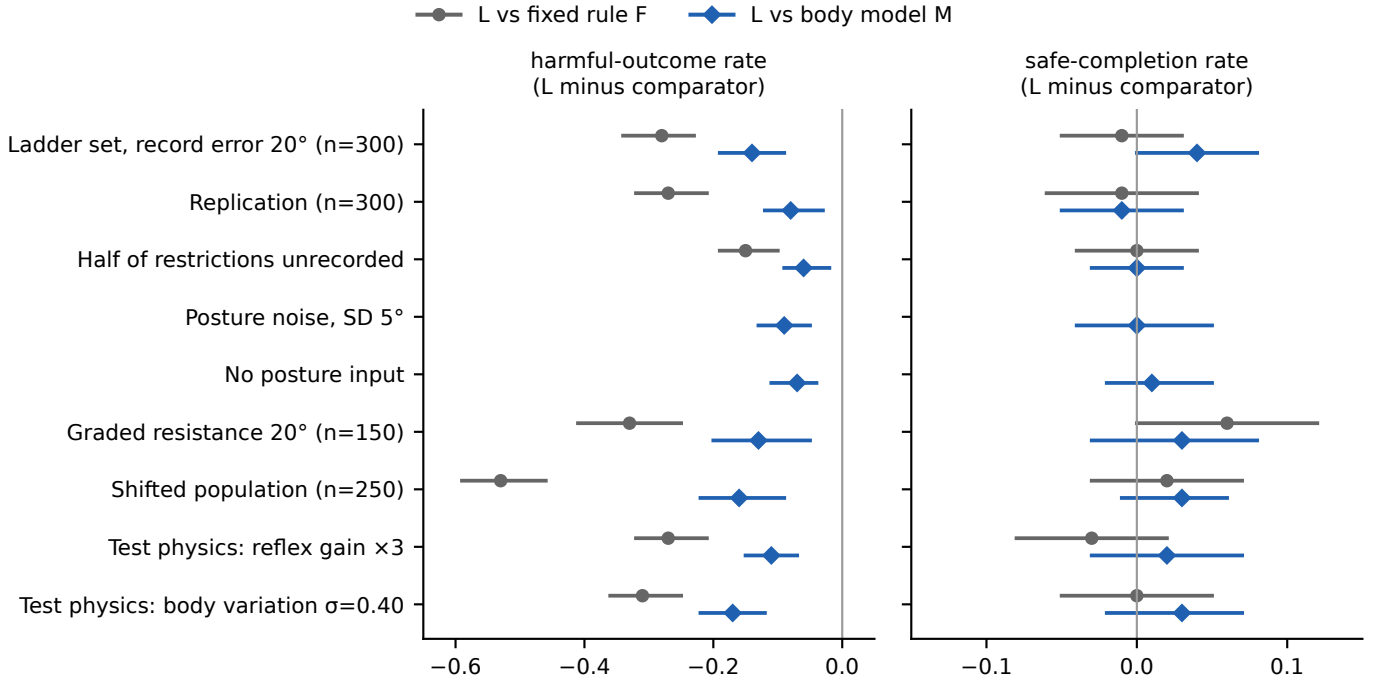


Fig. 4. The learned policy’s reduction in harm holds in every prespecified confirmation condition but shrinks when half of the restrictions are unrecorded. Paired differences (95% confidence intervals) in harmful-outcome rate (left) and safe-completion rate (right) between L and the fixed rule F (grey) or the body-model rule M (blue), harmful ≥ 0.5 N·m. The nine rows come from five independent person sets (rounds 24, 25A, 25B, 26 S and 26 V). L, M and F are not retrained; the no-posture variant is the separately specified four-feature policy. Posture noise is Gaussian with SD 5°; posture conditions are compared with M only, as prespecified. Counts are in Supplementary Table S5.

What kind of knowledge

The robot does not need a detailed personal model: probing adds harm, and record accuracy and even the recorded degrees matter little. What matters is knowing *which* joint is restricted, together with a policy learned from the simulated outcomes of many similar persons. M relies on a library of unrestricted bodies and must infer harm from predicted excursions; L learns harm labels from simulated persons of the restricted population and keeps its advantage in the shifted population.

The cost of declining

Declining hands the task to a human carer, whose handling carries its own risk (harm counts here are robot-caused); it mirrors the care rule to stop at resistance and report. L declines 30% of persons, the oracle 21%. Because τ keeps safe completions close to F’s, similar completion counts are partly a design choice (Supplementary Figures S1 and S2); a deployed system would set τ with the care team. For human carers, the assessment components of patient-handling programmes, which reduce staff injuries, have not been evaluated separately [45], and for falls a scored risk tool is no better than nurses’ judgement [46]; the ladder therefore describes robots, not carers. The simulated persons do not wince or speak; the results apply most directly to people who cannot give such cues, such as those with advanced dementia or reduced consciousness.

Implications for design

Robots that move limbs by end-effector contact should (i) obtain joint-level information before contact—at least a per-

joint, passive pass/fail screen for the range the task needs, repeated after significant change, which current merged-limb functional items do not provide; (ii) include an explicit option to decline and hand over the task; and (iii) not rely on force-magnitude thresholds to protect joints. The first two echo adjacent standards [13], [14], [16]; this study adds quantitative evidence.

Limitations and future work

All results are simulated, and training and test simulators share the musculoskeletal structure and joint-limit geometry; an arm phantom with adjustable limits and a change of body geometry between training and test are the next tests. The harm measure has no dose-response and depends on the simulator’s constraint model. The model restricts elbow flexion, not the clinically more common loss of extension, assumes independent restrictions with small uniform free ranges, and idealises the complete flag. The study covers one task, single-point support, a passive recipient and only force-magnitude guards; richer reactive controllers—estimating joint torques from the force vector with a limb model, detecting a stalled lift, compliant control, slower lifting—and two-point support at elbow and wrist, as carers use, may narrow the gap and deserve testing. The protocol was kept locally, and the analyses of what the robot needs to know are post hoc.

Conclusion

For a robot lifting a person’s arm, gentleness depends less on sensing the contact than on knowing, before con-

tact, which joint is restricted and when not to lift. Force-magnitude thresholds do not substitute for that knowledge, and unrecorded restrictions remain unprotected. With a complete record, a learned policy has about a quarter of a fixed rule's harmful outcomes, but its safe-completion non-inferiority is not established.

CONFLICT OF INTEREST STATEMENT

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

GENERATIVE AI STATEMENT

The author declares that generative AI was used in the creation of this manuscript: Claude (model Claude Opus 5.5, Anthropic, accessed through Claude Code in September 2026) was used to write and run simulation and analysis code, search and summarise literature, check numbers and references, and draft and edit the text, under the author's direction. The author reviewed and verified all code, results, references and text and takes full responsibility for the content.

AUTHOR CONTRIBUTIONS

MW: conceptualization, methodology, software, formal analysis, investigation, writing – original draft, writing – review and editing.

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SUPPLEMENTARY MATERIAL

Supplementary Table S1 lists every prespecified hypothesis with its outcome; Table S2 deviations and reuse of persons; Table S3 simulation parameters; Tables S4 and S5 the counts behind Figures 3 and 4; Table S6 the confirmation rounds. Supplementary Figure S1 shows the post hoc harm-threshold sweep and Figure S2 the post hoc trade-off between safe completions and harmful outcomes.

DATA AVAILABILITY STATEMENT

The simulation code, the dated protocol with all amendments, per-episode outcome data for every confirmation set, and scripts that regenerate every number, table and figure in this article are available at <https://github.com/maurice1128/gentle-by-knowing>.

ETHICS STATEMENT

This study used only computer simulation. No human participants, human data or animals were involved, and ethical approval was not required.

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